Analysis Techniques

Metadata and Data Carving

This set of tasks is designed to help you understand the role and importance of metadata in digital investigations, while also developing your knowledge of file signatures and the principles of file carving.

1. Metadata: EXIF Data

Digital photographs are an important source of evidence in digital forensics. Today, images can be easily created by witnesses or suspects using smartphones, digital cameras, and closed-circuit television (CCTV) systems. As a result, investigators frequently encounter digital photographs during forensic examinations.

Most digital cameras store photographs together with additional information known as metadata. This metadata is commonly stored using the Exchangeable Image File Format (EXIF). EXIF was developed by the Japanese Electronics Industry Development Association (JEIDA) to allow image-related information to be shared consistently between devices and software. EXIF data can include details such as the date and time a photo was taken, the make and model of the camera, and other technical settings.

Digital images can be saved in many different file formats, including JPEG, PNG, TIFF, BMP, GIF, PSD, JFIF, and others. Not all cameras or image formats use every available EXIF field, so the amount of metadata present can vary. However, even limited EXIF data can provide useful information for an investigation.

There are several tools available to examine EXIF metadata. One of the most commonly used tools is ExifTool. ExifTool is a command-line application that works across different operating systems and allows investigators to read, edit, and extract metadata from a wide range of file types. It is available for both Windows and macOS and is widely used in digital forensic practice.

1.1 EXIF header information

The information contained in the EXIF header can be used to focus searches it is possible to select some of the ASCII text in the EXIF header and use this as search criteria:

- Camera model and serial number: to determine if images originated on a particular / the same device.
- Flash Used - Images taken at night / indoors.
- Date and time picture was taken.
- GPS coordinates and GPS version ID
- Latitude, longitude, and altitude
- GPS timestamp
- Image description
- Author

1.2 Manipulating EXIF Headers

Several tools are available to assist someone manipulate the EXIF header – examples include: jhead, IMatch, PIE and others.

A. Research Task

Explore the above tools and their functionality.

Time for this task: 10 minutes

B. EXIF Data Exercise

A set of image files has been provided to you for this activity in the module's Teams channel in a folder labelled **EXIF**. Before initiating your analysis, transfer this folder into your forensic examination virtual machine (SANS SIFT) so that it is accessible within your working environment.

The analysis should be carried out using **ExifTool**. This utility is available within the SIFT workstation, but if it is not present in your setup, you should obtain it separately. Make use of the official ExifTool documentation throughout the exercise to support your analysis and ensure correct use of the tool.

1. Try to extract the metadata from the photos you have been provided by using the command:
   `exiftool <file_name>`
2. Select one of the photos and attempt to remove all metadata by using the command:
   `exiftool -all= <file_name>`
   Verify the type of metadata that has been removed from the file. What do you notice?
3. Attempt some metadata manipulation by
   a. renaming a photo with the Creation Date:
   `exiftool "-FileName<CreateDate" -d "%Y%m%d_%H%M%S.%%e" DIR`
   b. adding your own tag on a photo:
   `exiftool -p '"$artist"' -r DIR`
   c. changing the timestamp on a photo by 2 years:
   `exiftool "-DateTimeOriginal+=2:0:0 0" DIR`
4. Locate a media file supported by ExifTool, such as a .pdf or a .gif file. Then, attempt to create a text report to display the EXIF information. What information can you retrieve about this file?

Time for this task: 20 minutes

C. Optional Task - Metadata

For additional practice, you may choose to apply the tool to images you have personally captured.

2. File Carving

2.1 File Signatures

File signatures, often referred to as file headers, are specific sequences of bytes typically found at the beginning of a file. These signatures are used by operating systems and applications to recognise the file type and, in some cases, to interpret the file's content correctly.

File signatures are closely related to file extensions, which are the characters that appear after the final dot in a filename (for example, .jpg or .doc). However, file extensions can be easily changed by users and do not necessarily reflect the true nature of a file. For instance, a file originally named picture.jpg may be renamed to picture.doc in an attempt to disguise its contents.

Understanding file signatures is essential in digital forensics, as it allows investigators to identify the true nature of files regardless of their filename or extension. This is particularly important when dealing with deliberately disguised files, deleted data, or files recovered through carving, where traditional file system metadata may be missing or unreliable.

A. File Signatures Exercise

You are required to locate examples of the file types listed below and examine them using a hex editor of your choice.

By analysing the raw hexadecimal data, identify any file headers and footers present, and use your findings to complete the table provided.

File Extension	File header	File footer
.png	89 50 4E 47 - ‰PNG....	49 45 4E 44 AE 42 60 82 - IEND®B`,...
.img	50 49 43 54 00 08 - PICT..	none
.docx		
.doc		
.mov		
.mp3		
.zip		
.xlsx		
.jpg		

B. File Carving Exercise

When a file is only partially recovered, regardless of the file system, you can use file carving to attempt to recover the file. File carving is often used to recover data from a disk where there has been some damage or where the file itself is corrupt. This is a common method of data recovery, particularly when the file metadata has been damaged.

1. Whilst there are several automated tools available that can carry out file carving, the challenge in this practical is for you to carve a single file manually.
   Copy the file "file_carving_one.dd" to your machine. Use a hex editor of your choice to attempt to recover a JPG file from it.

Methodology:

- Identify the file header for the relevant file type you are interested in.
- Locate the file header in the hex editor (search).
- Determine the start and end of the file of interest.
- Select and copy the sectors in their correct sequence to a recovery file.
- Rebuild the file's header if necessary, to make it readable in a file viewer.
- Add the relevant file extension.

2. This file contains multiple image files, all in different formats. Download PhotoRec, read the associated documentation and run it against the file "file_carving_one.dd" to extract the other images.

3. Choose one of the images that was recovered by PhotoRec (not the JPG), confirm the file headers for that file type, and using the HEX editor, find that file within "file_carving_one.dd" to confirm the functionality of PhotoRec.

Note: PhotoRec has written a report that details the offset and length of the recovered file.

4. Confirm the MD5/SHA1 hashes for the JPG that you recovered and the JPG that PhotoRec recovered. Are they the same and if not, why?

Time for this task: 20 minutes

C. Optional Task – File Extensions

a. Research on what types of files the following extensions refer to and complete the table:

File Extension	File type
BMP	
LNK	
DLL	
ICS	
EXE	
OGG	
MOV	
JPG	

b. What program is used to open each one of the above file extensions?

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c. Can a user preview the content of these files?

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